DEVELOPMENT REPORT

Camera-Compensated Part-Trajectory Residual Development

Recording-grouped evaluation of residual motion and body-region experts

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Repository
https://github.com/abdullahuseyinli-dot/human-activity-classification

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Summary

This study investigates the remaining standing-versus-locomotion and transition errors in the Okutama-Action temporal classifier. The implementation keeps the established five-seed static and 0.5-second temporal models frozen, then adds bounded residual experts for center-conditioned temporal context, person-box kinematics, and confidence-masked body regions. Separate posture and motion gates decide how much of each residual reaches the factorized classifier.

The strongest development component combined the center-conditioned short-window encoder with seven body-region streams. Its five-seed probability ensemble improved macro-F1 from 0.7806 to 0.7887 on the fixed validation split, including a standing-F1 increase from 0.7023 to 0.7243. That result did not carry across the full grouped cross-fit: out-of-fold macro-F1 was 0.7144 for the new component and 0.7165 for the matched temporal baseline. The existing temporal ensemble therefore remains the default model, while the center-and-parts branch remains available for further work.

This distinction matters. A three-recording validation split showed a consistent positive direction, but the 11-recording OOF evaluation exposed seed and scenario sensitivity that a single split could not reveal.

1. Question and evaluation boundary

The experiment asks whether explicit motion structure can improve the classes that remain difficult after the v3 temporal study without weakening its static fallback. The design covers five mechanisms:

1. remove camera motion from person-box trajectories;
2. query short clips from their labeled center frame rather than globally pooling;
3. represent coarse body regions as confidence-weighted temporal tokens;
4. constrain learned evidence with motion-null, order, visibility, and reliability signals;
5. retain the frozen temporal model whenever a new component does not pass the fixed development gates.

All label-bearing work uses the Okutama provider-train archive. The locked development manifest contains 8,339 samples partitioned by recording: 4,977 training samples from 11 recordings, 1,383 validation samples from three recordings, and 1,979 calibration samples held closed during this study. Synchronized drone views stay in the same group. The provider test archive used by the earlier v3 confirmation is not opened or rescored.

The primary metric is three-class macro-F1 over sitting, standing, and combined walking/running. Accuracy, log loss, calibration error, per-class F1, transition subgroups, visibility subgroups, and recording-level results are retained alongside it.

2. Model

2.1 Frozen anchors

Each run loads the matching-seed v3 static model and 8-frame, 0.5-second temporal teacher. Their parameters remain frozen. The static model supplies factorized posture and motion logits, while the temporal teacher supplies the established short-clip residual. The new experts can add bounded residual evidence but do not silently replace either anchor.

The frame representation is the revision-pinned `facebook/dinov2-base` checkpoint. For each tracked person and frame, the cache contains a tight person crop, a context crop, and six normalized box-geometry values. All candidate training ran with CUDA automatic mixed precision on an NVIDIA GeForce RTX 4060 Laptop GPU.

2.2 Center-conditioned temporal encoder

The short encoder receives eight samples spanning 0.5 seconds and always includes the labeled center frame. A signed temporal encoding makes position relative to that center explicit. Two pre-norm Transformer layers encode the sequence, and the center token queries the encoded window through multi-head attention. This differs from unconditional sequence pooling: the prediction is explicitly about the tracked person at the labeled instant.

An independently instantiated long-clock encoder was evaluated over 1.0 second. Both shared-role and posture/motion-specialized variants were tested. Neither improved the short encoder, so the selected component uses only the short clock.

2.3 Camera-compensated kinematics

The trajectory path constructs a 21-dimensional signal per frame and a 58-dimensional summary. It includes normalized box center, scale and aspect, first and second temporal derivatives, path length, net displacement, straightness, camera transform diagnostics, validity, and camera-estimation quality.

Background-dominant sparse Lucas-Kanade tracks estimate a partial affine transform to the center frame. RANSAC rejects inconsistent tracks; phase-correlation translation is the declared fallback. Raw trajectories remain available whenever compensation is unreliable. The cache covers all 8,339 development samples and records a mean camera quality of 0.9908.

The recorded seed-42 trajectory screen moved from 0.7709 with raw box motion to 0.7723 with camera compensation. A later screen that added the compensated path to the center-conditioned encoder was 0.0026 higher than its recorded reference, just below the fixed 0.003 component gate. The integrated center, part, and trajectory screen reached 0.7678, so trajectory features were retained as a control rather than included in the five-seed candidate. These sequential screens span several implementation snapshots and are used to document candidate development, not to estimate matched causal effects.

2.4 Confidence-masked body-region encoder

Seven coarse regions are extracted from every tracked person crop: head/shoulders, torso, pelvis, left and right upper sides, and left and right legs. Each region carries a frozen DINOv2 token and a confidence derived from visibility and crop support. Within each frame, a spatial Transformer pools only supported regions. A second center-query temporal encoder then aggregates the part sequence. The expert's reliability is the joint temporal-validity and part-confidence mean.

The full part cache has shape 8,339 x 17 x 7 x 768 and a mean confidence of 0.4771. The subsequent body-region screen reached 0.7775 macro-F1, 0.0031 above the recorded center-only reference. This comparison selected the branch for the matched five-seed evaluation; it is part of the sequential development trace described below.

2.5 Gated residual fusion

Every learned expert produces separate two-logit posture and motion residuals. A quality-conditioned gate receives frozen static features, static probability diagnostics, center and window occlusion, box geometry, camera quality, part quality, and trajectory straightness. Separate sigmoid gates control posture and motion paths. Expert-role masks allow a modality to serve posture, motion, or both.

Residual heads are zero-initialized. Gate biases strongly favor the legacy temporal path at initialization. During training, quality-conditioned modality dropout and stochastic depth remove unreliable learned paths more often, while the frozen legacy path remains available. The selected model has 4.11 million parameters, of which 2.15 million are trainable.

2.6 Auxiliary and counterfactual objectives

The primary loss factorizes sitting/upright posture from stationary/locomoting motion. Auxiliary heads estimate transition windows, walking-versus-running subtype when the provider label supports it, and window visibility. The declared counterfactuals are:

- repeat the center frame and zero relative motion;

- reverse temporal order for non-transition consistency;
- apply coherent camera-coordinate jitter;
- mask low-confidence part observations.

The first counterfactual formulation weakened the aggregate result. A refined version penalized only the learned residual under motion nulling and preserved the frozen legacy response. It raised the seed-42 transition subgroup from 0.5236 to 0.5549 macro-F1, but its overall macro-F1 was 0.7694 versus 0.7775 for the simpler model. The transition effect is retained as a follow-up result, not merged into the selected classifier.

2.7 Adaptation and robustness controls

The study also evaluates:

- label-free masked reconstruction of frozen target-video features with a temporal

order objective;

- a frozen center-frame SigLIP posture specialist;
- top-two-block DINOv2 query/value LoRA with 53,764 trainable parameters;
- scenario-weighted GroupDRO;
- a continuous utility router and the required static-confidence heuristics.

Masked pretraining converged, but its downstream macro-F1 was 0.7712. SigLIP reached 0.7720, GroupDRO 0.7752, and the LoRA specialist 0.7187. The learned router did not predict continuous true-class gain reliably; its strongest mandatory control was the inverse static-margin rule. None displaced the center-and-parts component.

The model and dataset interfaces also support a confidence-aware keypoint stream. No complete measured pose stream was available across the development split, so the optional pose input remained disabled.

3. Execution sequence

Stage	Seed-42 validation macro-F1	Recorded comparison	Decision
Frozen temporal reference	0.7489	-	Reference
Raw trajectory	0.7709	+0.0219 vs reference	Camera-control input
Compensated trajectory	0.7723	+0.0014 vs raw	Measured control
Center-conditioned short residual	0.7744	+0.0254 vs reference	Retained
Shared dual clock	0.7672	-0.0072 vs short	Rejected
Specialized dual clock	0.7681	-0.0062 vs short	Rejected
Short plus trajectory	0.7770	+0.0026 vs short	Below component gate
Short plus body regions	0.7775	+0.0031 vs short	Five-seed candidate
Short, regions, and trajectory	0.7678	-0.0097 vs short plus regions	Rejected
Refined counterfactual objective	0.7694	-0.0081	Transition diagnostic
Masked initialization	0.7712	-0.0062	Rejected
SigLIP posture specialist	0.7720	-0.0055	Rejected
GroupDRO	0.7752	-0.0023	Rejected
Top-block LoRA	0.7187	-0.0302 vs reference	Rejected

All branch outcomes and local artifacts were retained in the declared execution order. The component table is a sequential development trace: its rows do not all share the same model, feature, and training source revisions. The portable `component_ablation.csv` records each summary, request, and source hash so that this history remains inspectable. The five-seed promotion comparison and all 25 grouped cross-fit runs use one consistent model, feature, runner, grid, and data-store snapshot; those matched evaluations, rather than the early screens, determine the final decision.

4. Five-seed and grouped results

4.1 Fixed validation split

Model	Macro-F1	Accuracy	Log loss	Sitting F1	Standing F1	Locomotion F1
Frozen temporal ensemble	0.7806	0.7780	0.5632	0.8232	0.7023	0.8163
Center and parts ensemble	0.7887	0.7867	0.5576	0.8204	0.7243	0.8213
Change	+0.0081	+0.0087	-0.0056	-0.0029	+0.0220	+0.0050

The three validation recordings all have positive macro-F1 deltas: +0.0069, +0.0055, and +0.0197. With only three exchangeable groups, however, the exact paired recording-swap test has eight permutations and a two-sided p-value of 0.25. The narrow cluster-bootstrap interval should therefore not be interpreted without the exact test and the broader cross-fit below.

4.2 Recording-grouped cross-fit

For each of five folds, all samples from a recording stay together. Each fold is run with seeds 42 through 46 using fixed epoch counts selected before cross-fit. The 25 models cover every one of the 4,977 training rows exactly once per seed.

Model	Macro-F1	Accuracy	Log loss	Sitting F1	Standing F1	Locomotion F1
Frozen temporal ensemble	0.7165	0.7131	0.7539	0.7368	0.6767	0.7360
Center and parts ensemble	0.7144	0.7123	0.7503	0.7295	0.6773	0.7366
Change	-0.0020	-0.0008	-0.0035	-0.0073	+0.0006	+0.0006

The 95% paired recording-cluster interval for the macro-F1 change is [-0.0063, +0.0024], with bootstrap p = 0.396. The exact 2,048-permutation recording-swap test gives p = 0.471. Recording deltas range from -0.0136 to +0.0098. These results do not support replacing the frozen temporal ensemble.

5. Faithfulness and failure analysis

The five-seed validation ensemble was evaluated under interventions without refitting:

Intervention	Macro-F1	Change from real sequence	Prediction changes
Real sequence	0.7887	0.0000	0.0%
Repeat center / motion null	0.7446	-0.0441	12.5%
Remove part stream	0.7810	-0.0076	1.4%
Reverse temporal order	0.7860	-0.0027	2.2%
Deterministic temporal shuffle	0.7891	+0.0005	2.1%
Zero geometry	0.7879	-0.0008	0.6%
Geometry only	0.1789	-0.6098	66.7%
Coherent camera jitter	0.7887	0.0000	0.0%

Motion nulling reduces mean true-class log probability by 0.1140 and changes 12.5% of predictions. The real sequence rescues 107 cases and harms 43 relative to that counterfactual. Among the evaluated interventions, temporal appearance has the largest measured effect. Part tokens add a smaller validation contribution, while explicit geometry and camera jitter have little direct effect in the selected candidate.

The mean learned gates also remain conservative. Legacy gate means are 0.975 for posture and 0.932 for motion; center-short means are 0.223 and 0.267; part means are 0.084 and 0.133. Mean part reliability is 0.509.

Visibility is the largest failure mode observed in this evaluation. On clear OOF windows, the candidate is +0.0015 macro-F1 above baseline. On windows containing occlusion it is -0.0308 below. The validation split contains only 25 occluded windows, compared with 412 in OOF, so its aggregate gain underrepresents this failure mode. This pattern is more consistent with visibility-dependent instability than with transition status: the OOF transition delta is -0.0021, close to the overall delta.

6. Decision

The fixed promotion rule required at least +0.010 validation macro-F1, a positive paired cluster interval, no per-class regression larger than 0.010, and no material worst-recording regression. The candidate gains +0.0081 on validation, falls -0.0020 in grouped OOF, and has a -0.0136 worst-recording delta. It does not pass the aggregate, cross-fit, or worst-recording gates.

The development lock therefore keeps `v3_temporal_8f_050s_five_seed_ensemble` as the default. Calibration remains unopened. The strongest new component and all failed controls remain retained and auditable, with the occlusion-conditioned failure providing a specific target for the next iteration.

The retained cross-fit requests bind the model, feature module, runner, candidate grid, manifest, feature stores, and baseline predictions. Their original request schema did not include the shared training helper, and the original plan-lock schema did not bind the candidate grid or its adaptive-grid lock. A post-run contract audit verified that all 25 requests share the same recorded source hashes and that their grid hash matches the retained adaptive-grid lock. Current lock and runner schemas include the omitted bindings for future runs; no historical artifact or reported metric was rewritten.

7. Reproduction

Create the environment described in the repository README and use a CUDA-enabled PyTorch build. The following commands verify the retained final evidence after the feature caches and component screens have been produced:

```
# Verify the retained evidence without changing locks or completed runs.
.\.venv\Scripts\python.exe tools\lock_okutama_cptr_adaptive_grid.py --check
.\.venv\Scripts\python.exe tools\lock_okutama_cptr_stage2_grid.py --check
.\.venv\Scripts\python.exe tools\lock_okutama_cptr_stage3_grid.py --check
.\.venv\Scripts\python.exe tools\lock_okutama_cptr_stage4_grid.py --check
.\.venv\Scripts\python.exe tools\lock_okutama_cptr_crossfit_plan.py --check
.\.venv\Scripts\python.exe tools\lock_okutama_cptr_development.py --check

# Rebuild the portable export from retained runs, then validate the repository.
.\.venv\Scripts\python.exe tools\export_okutama_cptr_results.py
.\.venv\Scripts\python.exe tools\validate_repository.py
```

The complete protocol is in `experiments/okutama_cptr_protocol.json`. The portable tables and hashes are in `results/okutama_cptr`. Large caches and checkpoints stay in ignored local run directories.

References

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